

Task 1: To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior

```
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <limits.h>

int main() {
    char cwd[PATH_MAX];

    if (getcwd(cwd, sizeof(cwd)) != NULL) {
        printf("Current Directory: %s\n", cwd);
    } else {
        perror("getcwd");
        return 1;
    }

    printf("Process ID: %d\n", getpid());

    printf("Type 'exit' to terminate.\n");

    char command[50];

    while (1) {
        printf("shell> ");
        scanf("%49s", command);

        if (command[0] == 'p' &&
```

```

    if (command[0] == 'p' &&
        command[1] == 'w' &&
        command[2] == 'd') {

        if (getcwd(cwd, sizeof(cwd)) != NULL)
            printf("%s\n", cwd);

    } else if (command[0] == 'e' &&
               command[1] == 'x' &&
               command[2] == 'i' &&
               command[3] == 't') {

        printf("Exiting...\n");
        break;

    } else {
        printf("Invalid command\n");
    }
}

printf("Resources cleaned up. Program terminated.\n");

return 0;
}

```

```

shreyas@Grigori:~$ mkdir -p ~/skill_session_10
shreyas@Grigori:~$ cd ~/skill_session_10
shreyas@Grigori:~/skill_session_10$ nano pwd_exit.c
shreyas@Grigori:~/skill_session_10$ gcc pwd_exit.c -o pwd_exit
shreyas@Grigori:~/skill_session_10$ ./pwd_exit
Current Directory: /home/shreyas/skill_session_10
Process ID: 545
Type 'exit' to terminate.
shell> pwd
/home/shreyas/skill_session_10
shell> abc
Invalid command
shell> pwd
/home/shreyas/skill_session_10
shell> exit
Exiting...
Resources cleaned up. Program terminated.

```

Task 2: To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality

```
GNU nano 7.2 export_test.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char input[200];

    while (1) {
        printf("shell> ");
        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (strncmp(input, "export ", 7) == 0) {
            char *assignment = input + 7;
            char *equal = strchr(assignment, '=');

            if (equal == NULL) {
```

```
                if (equal == NULL) {
                    printf("Invalid export syntax\n");
                    continue;
                }

                *equal = '\0';

                char *name = assignment;
                char *value = equal + 1;

                if (setenv(name, value, 1) == 0)
                    printf("%s=%s\n", name, value);
                else
                    perror("export");
            } else if (strncmp(input, "echo $", 6) == 0) {
                char *name = input + 6;
                char *value = getenv(name);

                if (value != NULL)
                    printf("%s\n", value);
                else
                    printf("Variable not found\n");
            } else if (strcmp(input, "test-child") == 0) {
```

```
GNU nano 7.2 export_test.c *
if (pid == 0) {
    char *value = getenv("MYVAR");
    if (value != NULL)
        printf("Child sees MYVAR=%s\n", value);
    else
        printf("Child cannot find MYVAR\n");
    exit(0);
} else if (pid > 0) {
    wait(NULL);
    printf("Child completed\n");
} else {
    perror("fork");
}
} else {
    printf("Invalid command\n");
}
}
return 0;
```

```
shreyas@Grigori:~$ mkdir -p ~/skill_session_10
shreyas@Grigori:~$ cd ~/skill_session_10
shreyas@Grigori:~/skill_session_10$ nano pwd_exit.c
shreyas@Grigori:~/skill_session_10$ gcc pwd_exit.c -o pwd_exit
shreyas@Grigori:~/skill_session_10$ ./pwd_exit
Current Directory: /home/shreyas/skill_session_10
Process ID: 545
Type 'exit' to terminate.
shell> pwd
/home/shreyas/skill_session_10
shell> abc
Invalid command
shell> pwd
/home/shreyas/skill_session_10
shell> exit
Exiting...
Resources cleaned up. Program terminated.
shreyas@Grigori:~/skill_session_10$ nano export_test.c
shreyas@Grigori:~/skill_session_10$ gcc export_test.c -o export_test
shreyas@Grigori:~/skill_session_10$ ./export_test
shell> export MYVAR=Shreyas
MYVAR=Shreyas
shell> echo $MYVAR
Shreyas
shell> test-child
Child sees MYVAR=Shreyas
Child completed
shell> exit
shreyas@Grigori:~/skill_session_10$ |
```

```

shreyas@Grigori:~/skill_session_10$ echo $HOME
/home/shreyas
shreyas@Grigori:~/skill_session_10$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Python314/Scripts/:/mnt/c/Python314/:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Users/shrey/Downloads/jdk-25_windows-x64_bin/jdk-25/bin:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Program Files/PuTTY:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/shrey/.local/bin:/mnt/c/Users/shrey/Downloads/flutter_windows_3.35.6-stable/flutter/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Python313/Scripts:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Python313:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/shrey/AppData/Local/Microsoft/WindowsApps:/mnt/c/wsl64/ucrt64/bin:/mnt/c/Program Files/JetBrains/CLion 2025.2/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/JetBrains/IntelliJ IDEA Community Edition 2025.2.3/bin:/mnt/c/Users/shrey/AppData/Local/JetBrains/Toolbox/scripts:/mnt/c/Program Files/JetBrains/WebStorm 2025.3.2/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Antigravity/bin:/mnt/c/Program Files/JetBrains/PyCharm 2025.2.4/bin:/mnt/c/Users/shrey/AppData/Roaming/npm:/mnt/c/Users/shrey/AppData/Local/Python/bin:/mnt/c/Users/shrey/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Users/shrey/Downloads/pypy3.12-v8.0.0-win64/snap/bin
shreyas@Grigori:~/skill_session_10$ export TESTVAR=Hello
shreyas@Grigori:~/skill_session_10$ echo $TESTVAR
Hello
shreyas@Grigori:~/skill_session_10$ |

```